

Synopsis

Traffic Accident Severity Prediction Using Machine Learning

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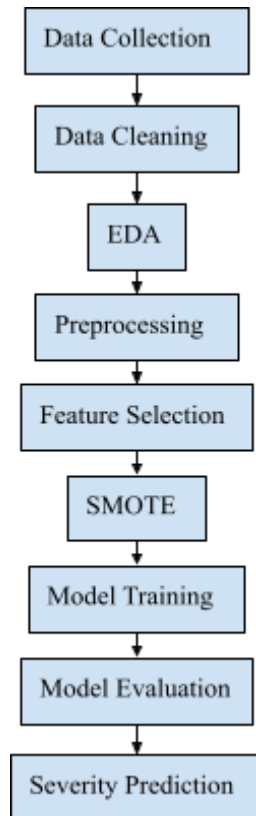
Problem Description

Traffic accidents are a major issue affecting road safety and public health. The severity of accidents depends on several factors such as vehicle speed, weather conditions, road type, and vehicle category. Predicting accident severity can help authorities and emergency services respond more efficiently and reduce damage and casualties.

This project aims to develop a machine learning system that predicts the severity of traffic accidents based on various input features. The system will use historical accident data and apply preprocessing techniques to clean the dataset and remove irrelevant records. Feature selection methods will be used to identify the most important factors affecting accident severity, such as speed and weather conditions.

Machine learning algorithms such as Logistic Regression and Support Vector Machine (SVM) will be used to build the prediction model. The SVM model will use different kernels such as Linear, Polynomial, and RBF to capture simple and complex patterns in the data. The performance of the model will be evaluated using metrics like confusion matrix and recall score to measure prediction accuracy.

Architecture



Dataset Collection

The road accident dataset was collected containing driver, vehicle, road, weather, and accident-related information.

Data Cleaning

Missing values, duplicate entries, and inconsistent categories were handled to improve data quality.

EDA (Exploratory Data Analysis)

The dataset was analyzed using graphs and visualizations to identify patterns, class imbalance, and important features.

Preprocessing

Categorical data was encoded and numerical data was scaled for machine learning model compatibility.

Feature Selection

Important features affecting accident severity were selected to reduce dimensionality and improve performance.

SMOTE

SMOTE was applied to balance the dataset by increasing minority class samples.

Model Training

Different SVM kernels (Linear, RBF, Polynomial, Sigmoid) and Logistic Regression were trained and tested.

Model Evaluation

Models were evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

Severity Prediction

The final system predicts whether an accident will result in slight or severe injury.

Proposed Modules

1. Data Collection Module

This module is responsible for loading and managing the road accident dataset used for analysis and prediction.

2. Data Cleaning Module

This module handles missing values, duplicate records, inconsistent categories, and category reduction to improve dataset quality.

3. Exploratory Data Analysis (EDA) Module

This module performs graphical and statistical analysis of the dataset using plots and visualizations to identify patterns and relationships between features.

4. Data Preprocessing Module

This module performs encoding, feature scaling, feature selection, and dataset balancing using SMOTE to prepare data for machine learning models.

5. Model Training Module

This module trains different machine learning models including Support Vector Machine and Logistic Regression for accident severity prediction.

6. Model Evaluation Module

This module evaluates model performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

7. Prediction Module

This module takes accident-related input data and predicts whether the accident severity will be slight injury or severe injury using the trained model.

8. Model Saving and Loading Module

This module saves trained models and preprocessing objects using joblib and reloads them during implementation and prediction.

Required Topics from the Subject

1. Python Programming

Python was used as the primary programming language for implementing data preprocessing, machine learning models, visualization, and prediction modules. Its simple syntax and extensive libraries made development easier and efficient.

2. File Handling

File handling was used to read datasets from CSV files and save trained machine learning models using pickle/joblib files. It enabled persistent storage and reuse of trained models during implementation.

3. Data Structures

Data structures such as lists, dictionaries, arrays, and dataframes were used for storing and manipulating accident-related data during preprocessing and model training.

4. Machine Learning

Machine learning concepts such as classification, feature selection, preprocessing, model training, and evaluation were used to predict accident severity using different algorithms.

5. Data Visualization

Visualization techniques were used to analyze dataset patterns and model performance using graphs such as heatmaps, count plots, boxplots, and confusion matrices.

6. Statistical Analysis

Basic statistical concepts such as correlation, distribution analysis, and class imbalance analysis were used during exploratory data analysis and preprocessing.

Platform Used

Jupyter Notebook

Jupyter Notebook was used for the complete development and implementation of the project. It provides an interactive environment for writing, testing, and executing Python code in separate cells. Jupyter was especially useful for data preprocessing, exploratory data analysis (EDA), visualization, machine learning model training, and result analysis. Its notebook-based structure made experimentation, debugging, and visualization easier throughout the project development process.

References

- Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow – Aurélien Géron
- Pattern Recognition and Machine Learning – Christopher Bishop
- <https://scikit-learn.org>
- <https://www.kaggle.com>